

BEOGRAD SURCIN, SerbiaWMO: **132720**Lat: **44.818N**Lon: **20.309E**Elev: **335**StdP: **14.52**Time Zone: **1.00 (EUC)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.8	18.0	8.7	8.7	15.9	13.9	11.3	20.2	25.6	37.2	22.9	37.1	6.6	110	0.470

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.8	94.7	70.6	91.3	70.5	88.0	69.2	73.6	88.0	72.2	86.0	70.7	83.9	7.7	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.5	109.8	79.3	67.9	103.7	77.8	66.4	98.3	75.8	37.3	88.1	36.0	86.1	34.8	83.9	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.2	19.0	16.4	6.3	99.3	7.9	3.8	0.6	102.0	-4.0	104.2	-8.5	106.3	-14.2	109.0		
(5)			5.7	76.3	7.9	1.6	0.0	77.5	-4.6	78.4	-9.0	79.3	-14.7	80.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.7	34.0	37.9	45.7	55.3	63.7	70.0	73.3	73.4	64.7	55.6	46.2	36.0		
	DBStd	15.98	8.90	9.81	8.61	7.49	6.85	6.71	6.01	6.07	6.91	7.97	8.81	8.75		
	HDD50	1726	500	350	186	36	1	0	0	0	0	38	176	439		
	HDD65	4627	963	759	599	300	109	28	7	7	90	302	565	898		
	CDD50	3454	3	11	52	194	425	602	723	726	440	211	61	6		
	CDD65	878	0	0	0	8	68	180	264	268	80	10	0	0		
	CDH74	9202	0	0	4	103	651	1858	2862	2874	744	104	2	0		
	CDH80	3693	0	0	0	15	151	721	1261	1306	229	10	0	0		
(14) Wind	WSAvg	6.9	7.1	7.9	8.2	7.9	6.8	6.2	6.0	5.7	6.2	6.6	7.4	7.0		
(15) Precipitation	PrecAvg	26.3	1.7	1.5	1.8	2.0	2.6	3.3	2.6	2.1	2.5	2.1	1.9	2.0		
	PrecMax	39.8	3.3	3.5	4.6	5.8	9.6	6.9	9.3	5.0	7.4	4.5	5.3	5.5		
	PrecMin	13.9	0.3	0.0	0.2	0.1	0.7	0.7	0.4	0.1	0.4	0.0	0.1	0.2		
	PrecStd	6.3	0.8	0.8	1.3	1.1	1.7	1.8	2.2	1.4	1.4	1.3	1.1	1.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.0	66.0	73.0	82.3	88.0	94.7	98.7	100.0	91.6	80.9	72.0	61.2		
		MCWB	50.6	52.7	56.2	62.6	66.8	70.8	69.9	70.8	67.7	64.9	58.9	52.7		
	2%	DB	53.6	60.7	68.4	76.9	84.1	91.1	94.6	95.0	86.1	77.0	67.6	56.7		
		MCWB	47.6	50.4	53.4	60.1	65.6	71.4	70.7	70.7	66.1	62.5	57.2	49.7		
	5%	DB	50.1	55.7	64.4	72.7	80.8	87.8	91.0	91.3	82.1	73.1	62.8	52.0		
		MCWB	45.6	47.9	52.1	57.6	64.9	69.8	70.8	70.0	65.3	60.7	54.5	46.9		
	10%	DB	46.4	51.8	60.1	68.8	77.4	84.3	87.4	87.6	78.4	68.4	59.1	48.1		
		MCWB	43.1	45.6	49.8	55.6	63.3	68.5	69.7	68.8	63.8	59.0	52.7	44.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.1	53.6	57.6	63.8	69.4	75.4	75.1	76.1	71.0	67.2	60.5	53.7		
		MCDB	57.1	63.7	69.8	78.2	81.7	88.3	90.1	89.4	86.2	78.5	69.6	60.4		
	2%	WB	48.4	50.9	55.0	61.2	67.3	72.9	73.3	73.3	68.4	63.9	57.5	50.2		
		MCDB	53.1	59.2	65.7	74.5	80.4	86.2	87.7	88.7	80.8	72.9	65.8	55.4		
	5%	WB	45.8	48.6	52.6	59.0	65.7	71.2	72.1	71.6	66.5	61.7	55.3	47.6		
		MCDB	49.4	55.1	62.4	70.3	78.4	84.1	86.0	85.3	77.8	70.6	61.8	51.9		
	10%	WB	43.1	46.1	50.4	56.8	64.0	69.6	70.7	70.1	64.8	59.8	53.0	44.7		
		MCDB	45.8	51.0	58.8	66.2	75.5	81.8	83.6	83.5	75.5	67.6	58.6	47.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.2	14.1	17.4	19.0	19.4	19.6	20.8	21.3	19.1	18.0	14.4	10.7		
		MCDBR	17.5	21.4	24.4	25.4	24.0	24.9	26.4	27.2	25.2	24.0	20.7	17.0		
		MCWBR	12.5	13.1	13.1	12.4	10.2	9.5	9.2	9.3	10.0	11.8	12.0	11.9		
		MCDBR	16.6	19.8	22.4	22.5	22.6	22.7	23.0	23.2	22.1	21.9	19.0	16.0		
	5% WB	MCWBR	12.1	12.5	12.6	11.6	10.3	9.6	9.1	9.2	10.1	11.7	11.5	11.6		
(40) Clear-Sky Solar Irradiance	taub		0.346	0.373	0.404	0.461	0.448	0.444	0.447	0.456	0.435	0.399	0.361	0.344		
	taud		2.334	2.246	2.210	2.061	2.161	2.212	2.200	2.175	2.201	2.296	2.357	2.351		
	Ebn at Noon		238	252	260	254	262	263	260	253	247	240	230	225		
	Edh at Noon		26	33	39	50	46	44	44	44	39	31	25	23		
(44) All-Sky Solar Radiation	RadAvg		394	654	1044	1436	1751	1943	1970	1748	1224	842	511	330		
	RadStd		54	87	125	146	167	181	150	166	134	93	43	49		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+1.28	N/A	N/A	N/A	N/A	N/A	-292	+225	N/A
(47) Regional (50 neighbors)		+1.21	N/A	N/A	N/A	+0.65	N/A	-209	+214	+113

Nomenclature: See separate page